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FINALTERM EXAMINATION

CS301- Data Structures

Question No: 1 (Marks: 1) - Please choose one

The arguments passed to a function should match in number, type and order with the parameters in the function definition.

► True
► False

Question No: 2 (Marks: 1) - Please choose one

If numbers 5, 222, 4, 48 are inserted in a queue, which one will be removed first?

► 48
► 4
► 222
► 5

Question No: 3 (Marks: 1) - Please choose one

Suppose currentNode refers to a node in a linked list (using the Node class with member variables called data and nextNode). What statement changes currentNode so that it refers to the next node?

► currentNode ++;

► currentNode = nextNode;

► currentNode += nextNode;

► currentNode = currentNode->nextNode;

Question No: 4 (Marks: 1) - Please choose one

A **Compound Data Structure** is the data structure which can have multiple data items of same type or of different types. Which of the following can be considered compound data structure?

► Arrays
► LinkLists
► Binary Search Trees
► All of the given options

Question No: 5 (Marks: 1) - Please choose one

Here is a small function definition:

```
void f(int i, int &k)
{
      i = 1;
      k = 2;
}
```

Suppose that a main program has two integer variables `x` and `y`, which are given the value 0. Then the main program calls `f(x,y);` What are the values of `x` and `y` after the function `f` finishes?

► Both `x` and `y` are still 0.

► `x` is now 1, but `y` is still 0.

► `x` is still 0, but `y` is now 2.

► `x` is now 1, and `y` is now 2.

Question No: 6 (Marks: 1) - Please choose one

The difference between a binary tree and a binary search tree is that ,

► a binary search tree has two children per node whereas a binary tree can have none, one, or two children per node

► in binary search tree nodes are inserted based on the values they contain
► in binary tree nodes are inserted based on the values they contain
► ***none of these***

Question No: 7 (Marks: 1) - Please choose one

Compiler uses which one of the following to evaluate a mathematical equation,

► Binary Tree
► Binary Search Tree
► **Parse Tree**

► AVL Tree

Question No: 8 (Marks: 1) - Please choose one

If there are 56 internal nodes in a binary tree then how many external nodes this binary tree will have?

► 54
► **55**
► 56
► 57

Question No: 9 (Marks: 1) - Please choose one

If there are 23 external nodes in a binary tree then what will be the no. of internal nodes in this binary tree?

► 23
► 24
► 21
► 22

Question No: 10 (Marks: 1) - Please choose one

Which of the following method is helpful in creating the heap at once?

► insert
► add
► update
► preculateDown (lecture #31 page 1)

Question No: 11 (Marks: 1) - Please choose one

The definition of Transitivity property is

► For all element x member of S, x R x
► For all elements x and y, x R y if and only if y R x
► For all elements x, y and z, if x R y and y R z then x R z
► For all elements w, x, y and z, if x R y and w R z then x R z

Question No: 12 (Marks: 1) - Please choose one

A binary tree of N nodes has _____.

► Log_{10} N levels
► **Log_2 N levels**
► N / 2 levels

► N x 2 levels

Question No: 13 (Marks: 1) - Please choose one

If there are N elements in an array then the number of maximum steps needed to find an element using Binary Search is _____ .

► N
► N^2
► $N\log_2 N$
► **$\log_2 N$**

Question No: 14 (Marks: 1) - Please choose one

Consider te following array
23 15 5 12 40 10 7
After the first pass of a particular algorithm, the array looks like
15 5 12 23 10 7 40
Name the algorithm used

► Heap sort
► Selection sort
► Insertion sort
► **Bubble sort**

Question No: 15 (Marks: 1) - Please choose one

If both pointers of the node in a binary tree are NULL then it will be a/an _____ .

► Inner node
► Leaf node
► Root node
► None of the given options

Question No: 16 (Marks: 1) - Please choose one

By using _____we avoid the recursive method of traversing a Tree, which makes use of stacks and consumes a lot of memory and time.

► Binary tree only
► Threaded binary tree
► Heap data structure

► Huffman encoding

Question No: 17 (Marks: 1) - Please choose one

A complete binary tree of height *3* has between _____ nodes.

► 8 to 14
► 8 to 15
► 8 to 16
► 8 to 17

Question No: 18 (Marks: 1) - Please choose one

Consider a min heap, represented by the following array:
3,4,6,7,5,10
After inserting a node with value 1.Which of the following is the updated min heap?

► 3,4,6,7,5,10,1
► 3,4,6,7,5,1,10
► 3,4,1,5,7,10,6
► 1,4,3,5,7,10,6 close to correct but correct ans is 1,4,3,7,5,10,6

Question No: 19 (Marks: 1) - Please choose one

Consider a min heap, represented by the following array:
10,30,20,70,40,50,80,60
After inserting a node with value 31.Which of the following is the updated min heap?

► 10,30,20,31,40,50,80,60,70

► 10,30,20,70,40,50,80,60,31

► 10,31,20,30,40,50,80,60,31

► 31,10,30,20,70,40,50,80,60

Question No: 20 (Marks: 1) - Please choose one

Which one of the following algorithms is most widely used due to its good average time,

► Bubble Sort
► Insertion Sort
► Quick Sort
► Merge Sort

Question No: 21 (Marks: 1) - Please choose one

Which of the following statement is correct about find(x) operation:

► A find(x) on element x is performed by returning exactly the same node that is found.

► A find(x) on element x is performed by returning the root of the tree containing x.

► A find(x) on element x is performed by returning the whole tree itself containing x.

► A find(x) on element x is performed by returning TRUE.

Question No: 22 (Marks: 1) - Please choose one

Which of the following statement is NOT correct about find operation:

► It is not a requirement that a find operation returns any specific name, just that finds on two elements return the same answer if and only if they are in the same set.

► One idea might be to use a tree to represent each set, since each element in a tree has the same root, thus the root can be used to name the set.

► Initially each set contains one element.

► Initially each set contains one element and it does not make sense to make a tree of one node only.

Question No: 23 (Marks: 1) - Please choose one

The following are statements related to queues.

(i) The last item to be added to a queue is the first item to be removed False statement
(ii) A queue is a structure in which both ends are not used False statement
(iii) The last element hasn't to wait until all elements preceding it on the queue are removed False statement
(iv) A queue is said to be a last-in-first-out list or LIFO data structure. False statement

Which of the above is/are related to normal queues?

► (iii) and (ii) only
► (i), (ii) and (iv) only
► (ii) and (iv) only
► None of the given options

Question No: 24 (Marks: 1) - Please choose one

The maximum number of external nodes (leaves) for a binary tree of height H is _____

► 2^H

► 2^H+1

► 2^H-1

► 2^H+2

Question No: 25 (Marks: 1) - Please choose one

In complete binary tree the bottom level is filled from _____

► Left to right

► Right to left

► Not filled at all

► None of the given options

Question No: 26 (Marks: 1) - Please choose one

We are given N items to build a heap , this can be done with _____ successive inserts.

► N-1
► N
► N+1
► N^2

Question No: 27 (Marks: 1) - Please choose one

Suppose we had a hash table whose hash function is “n % 12”, if the number 35 is already in the hash table, which of the following numbers would cause a collision?

► 144

► 145

► 143

► 148

Question No: 28 (Marks: 1) - Please choose one

Here is an array of ten integers:
5 3 8 9 1 7 0 2 6 4
The array after the FIRST iteration of the large loop in a selection sort (sorting from smallest to largest).

► 0 3 8 9 1 7 5 2 6 4

► 2 6 4 0 3 8 9 1 7 5

► 2 6 4 9 1 7 0 3 8 5

► 0 3 8 2 6 4 9 1 7 5

Question No: 29 (Marks: 1) - Please choose one

What requirement is placed on an array, so that *binary search* may be used to locate an entry? ► The array elements must form a heap.

► The array must have at least 2 entries.

► The array must be sorted.

► The array's size must be a power of two.

Question No: 30 (Marks: 1) - Please choose one

In case of deleting a node from AVL tree, rotation could be prolong to the *root* node.

► Yes

► **No**

Question No: 31 (Marks: 1)

In merge sort do we need to have extra memory, justify your answer in either case.
Yes we need extra memory in merge sort.

Question No: 32 (Marks: 1)

Where is Inorder Predecessor of a non leaf node is present in a Binary Search Tree?

Question No: 33 (Marks: 2)

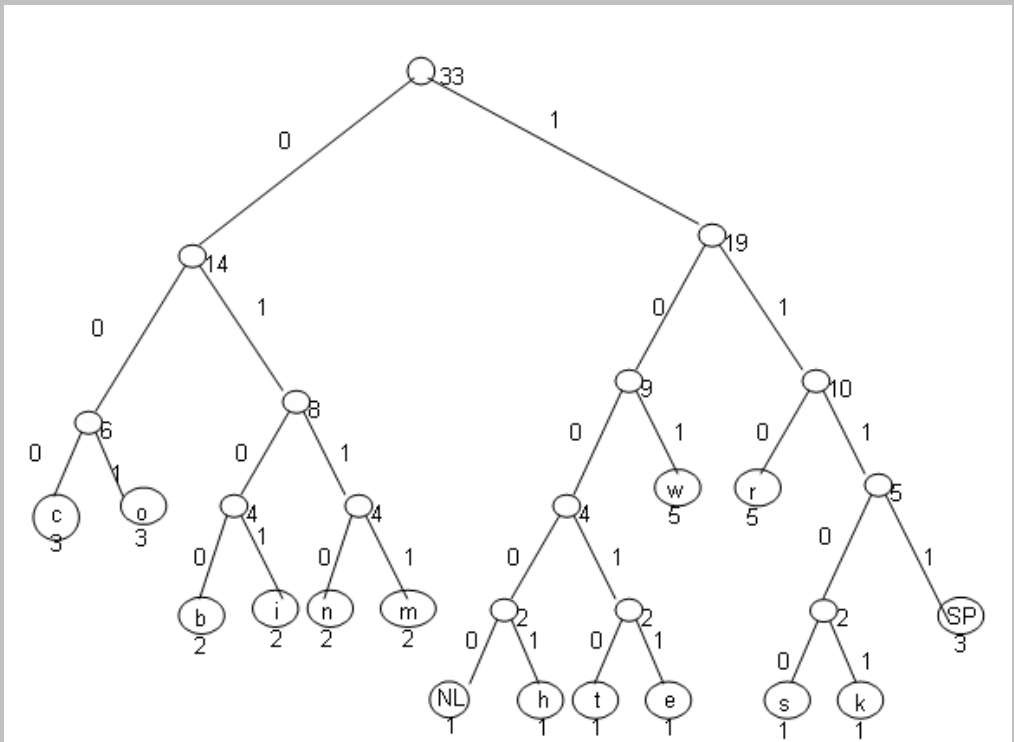
How we can search an element in Skip List.

Question No: 34 (Marks: 2)

What is the drawback of using arrays to store Binary Search Trees.

Question No: 35 (Marks: 3)

Calculate the codes of the following characters in table below using the hoffman encoding tree,



Hoffman encoding tree with path values

character	Code
NL	10000
SP	1111
o	001
b	0100
i	0101
r	110

Question No: 36 (Marks: 3)

"For smaller lists, linear insertion sort performs well, but for larger lists, quick sort is suitable to apply." Justify why?

Question No: 37 (Marks: 3)

Suppose that we have implemented a priority queue by storing the items in a heap. We are now executing a reheapification downward and the out-of-place node has priority of 42. The node's parent has a priority of 72, the left child has priority 52 and the node's right child has priority 62. Which statement best describes the status of the reheapification.

A. The reheapification is done.
B. The next step will swap the out-of-place node with its parent.
C. The next step will swap the out-of-place node with its left child.
D. The next step will swap the out-of-place node with its right child.
E. None of these.

Question No: 38 (Marks: 5)

Give two different reasons to explain why the following binary tree is not a heap:



Question No: 39 (Marks: 5)

Here is an array of ten integers:

5 3 8 9 1 7 0 2 6 4

Sort the array by using selection sort algorithm and show content of array after each step.

	5	3	8	9	1	7	0	2	6	4
STEP 1	0	3	8	9	1	7	5	2	6	4

STEP 2	0	1	8	9	3	7	5	2	6	4
STEP3	0	1	2	9	3	7	5	8	6	4
STEP 4	0	1	2	3	9	7	5	8	6	4
STEP5	0	1	2	3	4	7	5	8	6	9
STEP6	0	1	2	3	4	5	7	8	6	9
STEP 7	0	1	2	3	4	5	6	8	7	9
STEP8	0	1	2	3	4	5	6	7	8	9
	0	1	2	3	4	5	6	7	8	9

Question No: 40 (Marks: 10)

A long sequence of vowels needs to be transmitted efficiently so a programmer decides to use Huffman encoding to encode the vowels. A distribution count study of typical data yielded the following frequency table.

Frequency Table

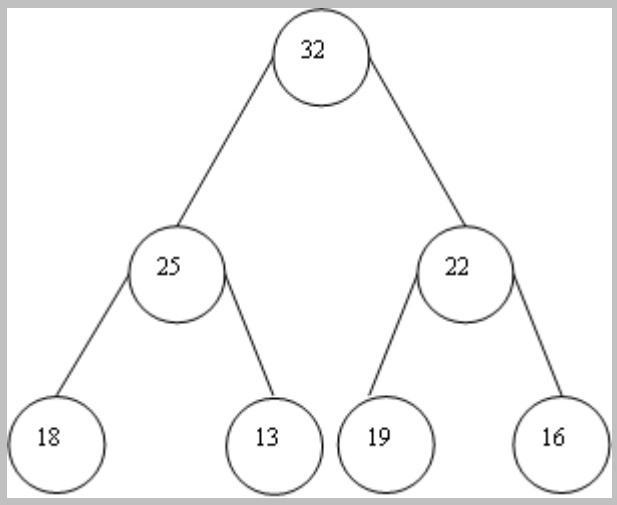
character	frequency	Huffman code
A	33978	_ _ _ _ _ _ _
E	20676	_ _ _ _ _ _ _
I	15814	_ _ _ _ _ _ _
O	21552	_ _ _ _ _ _ _
U	10324	_ _ _ _ _ _ _
Y	4975	_ _ _ _ _ _ _

A) Create a Huffman tree to determine the binary codes for each character.
B) Fill the codes into the table above.
C) Encode the following sequence EIYOUA

Question No: 41 (Marks: 10)

Consider the following tree.

a) Show that either it is a heap or not.
b) If it is a heap then what type of heap is it?
c) Add 40 in the heap and convert it in max heap.


32
25
22
18
13
19
16